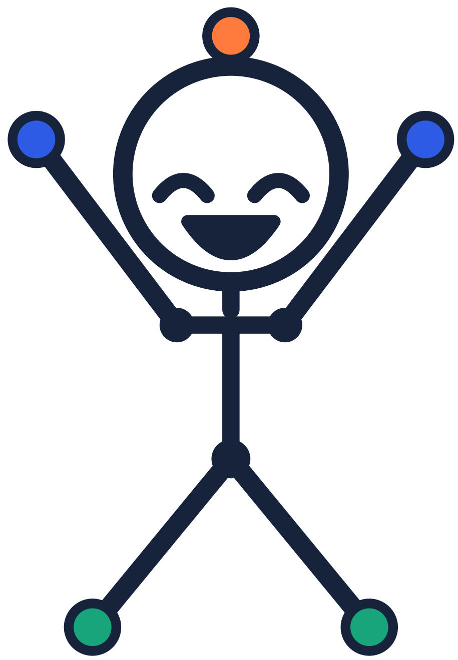
Young Innovators
FOR CHANGE
innovateyouth.org

A FULL-YEAR ENTERPRISE PROGRAM · GRADE 5

# The Numbers Year

36 weekly assignments. A partner, a line of products, and two selling sessions with one change in between.



**This book belongs to**

My partners

Our business

Teacher

School year

# Sell it. Measure it. Change one thing. Sell it again.

Two pages every week, front and back. About thirty-five minutes.

## The shape of a week

**Page 1** teaches one idea and gives you something to do with it. **Page 2** is practice, a question for home, and a harder challenge. It works as classwork or as homework, and it needs no computer.

Weeks 1 to 12 are the arithmetic, learned on other people's numbers. Weeks 13 to 24 you and a partner build a line of products and write a plan with a forecast in it. Weeks 25 to 36 you sell, measure, change one thing, and sell again.

## The boxes on the page

| | |
|---|---|
| **IN CLASS** | The Grade 5 standard your class is most likely working on that week. This book sits beside your math lesson rather than competing with it. |
| **MONEY WARM-UP** | Two minutes of arithmetic. Decimals this year, because a business is where 5.NBT.7 stops being an exercise. |
| **WORDS TO KEEP** | One or two words pinned down properly. Adults use these words loosely; you will not have to. |
| **NUMBERS FILE** | The one thing you build all year. Twenty-four pages by week 36, every one of them measured rather than remembered. |
| **FROM YOUR FILE** | Which earlier pages this week needs. This year is cumulative: week 22 is impossible without weeks 20 and 21. |
| **TALK ABOUT IT** | Two questions for someone at home. This is the part that makes the week stick, and it takes about five minutes. |
| **STUCK? TRY THIS** | A way in when the page has stopped making sense. Read it before you give up, not after. |
| **CHALLENGE ZONE** | Harder, and usually open. Where a student who finishes early should go. |

## About the two selling sessions

Weeks 29 and 34 are real selling. In between, the class works out what the numbers say and each team changes **one thing**. That is what makes this year an experiment rather than an event: whatever happens in session two, you know what caused it.

The two sessions do not need two separate days: two halves of one market with a break between works.

**The money is play money**, in a closed classroom economy. The class bank lends every team the same startup fund, they buy materials from a class supply table, sell and make their own change, then repay the fund. What is left is profit.

# Money to the Penny

Last year you worked in dollars. A real business lives in the cents after them.

**IN CLASS THIS WEEK**
Adding and subtracting decimals to hundredths (5.NBT.7)

**MONEY WARM-UP · 2 MINUTES**
A cup costs $0.37 to make. You make 24 of them. What did they cost altogether? ________

**WORDS TO KEEP**

| | |
|---|---|
| **to the penny** | Written with both decimal places, because the cents add up. |
| **rounding** | Fine for a rough guess. Never for a cost or a total. |

$ 0 . 3 7

dollars dimes pennies

## Why the cents matter

Thirty-seven cents does not feel like a number worth worrying about. Over 200 sales it is **$74**.

Beginners round to the nearest dollar because it is easier. Then the total is wrong by more than the profit was. This year every cost, every price and every total is written to the penny.

## Round or exact?

| Situation | Round or exact? | Why |
|---|---|---|
| Guessing if an idea is worth trying | round | you only need roughly |
| Working out cost per item | | |
| Telling a customer the price | | |
| Counting the money box at the end | | |

## Line them up

Decimal points under decimal points, always. Work these out.

| | Problem | Answer |
|---|---|---|
| a | $0.37 + $1.20 + $0.08 | |
| b | $5.00 − $3.65 | |
| c | $0.45 × 12 | |
| d | $14.40 ÷ 12 | |
| e | $2.50 + $0.99 + $0.07 | |

## The rounding trap

A seller says each item costs “about 50 cents” when it really costs $0.62. They make 150. How far out is their total?

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**▶ ADD TO YOUR NUMBERS FILE**

Start your Numbers File. Write the date on the front. Everything you work out this year goes in it, and by June it is the record of a real business.

**💬 TALK ABOUT IT**

Ask an adult to find something at home with a price ending in 99 cents.

Ask why they think the shop chose that number instead of a round one.

**STUCK? TRY THIS**

Write the numbers in a column with the decimal points lined up before you add. Most decimal mistakes are lining-up mistakes, not arithmetic mistakes.

**★ CHALLENGE ZONE**

A shop sells 1,000 items a week and its cost per item is wrong by 3 cents.

Work out how much that mistake costs over a whole year. Then say whether 3 cents is a small number.

WEEK 2 MEASURING

# Two Kinds of Cost

Some costs happen once. Some happen again every single time you sell one.

**IN CLASS THIS WEEK**
Multiplying decimals (5.NBT.7)

**MONEY WARM-UP · 2 MINUTES**
A sign costs $9.00 once. Each cup costs $0.40. What do 20 cups plus the sign cost? ________

**WORDS TO KEEP**

| | |
|---|---|
| **fixed cost** | Paid once whatever happens. The sign, the table, the printing. |
| **variable cost** | Paid again for every single item you make. |


## Fixed and variable

A **fixed cost** is paid once whatever happens. The sign. The table. The stamp for the labels.

A **variable cost** happens again for every single item. The paper cup. The lemon. The bag it goes in.

Last year you added them all together. Splitting them is what lets you answer the question in week 5.

## Sort these

| Cost | Fixed or variable? |
|---|---|
| The cardboard for your sign | |
| Beads for one bracelet | |
| Borrowing a table for the day | |
| The bag each item goes in | |
| Printing 40 price labels at once | |

## Total cost at three sizes

Fixed costs $9.00. Variable costs $0.40 each.

| If you make | Variable cost | Fixed cost | Total |
|---|---|---|---|
| 10 | | $9.00 | |
| 25 | | $9.00 | |
| 50 | | $9.00 | |

Now work out the cost of **one** item at each size, fixed cost included.

| If you make | Cost of one, all in |
|---|---|
| 10 | |
| 25 | |
| 50 | |

## What you noticed

The cost of one went down as you made more. Say why in one sentence.

___

**💬 TALK ABOUT IT**

Ask an adult to name a cost their workplace pays whether it is busy or quiet.

Ask what a busy day costs them that a quiet day does not.

**STUCK? TRY THIS**

Ask of each cost: if I made one more, would I pay this again? If yes it is variable. If no it is fixed.

**★ CHALLENGE ZONE**

Making more spreads the fixed cost over more items, so each one gets cheaper.

So why does no business just make an enormous number of everything? Give two real reasons. Both of them are about what happens to the ones you do not sell.

WEEK 3 MEASURING

# Cost Per Unit, Properly

Divide to get one. Then remember the ones that went wrong.

**IN CLASS THIS WEEK**
Dividing decimals and whole numbers (5.NBT.6, 5.NBT.7)

**MONEY WARM-UP · 2 MINUTES**
A pack of 45 stickers costs $6.75. What does one sticker cost? ________

**WORDS TO KEEP**

**unit cost** What one item costs you, materials divided down and waste added on.

**spoilage** The ones that come out wrong. You paid for them anyway.

## Working down to one

Materials arrive in packs. You never use a whole pack on one item, so every material has to be divided down before it can be added up.

Worked example. A $6.75 pack holds 45 stickers, so one sticker is $6.75 ÷ 45 = **$0.15**. A card uses 3 stickers, so the stickers in one card cost $0.45.

## Do the same for these

| Material | Pack price | How many in a pack | Used per item | Cost per item |
|---|---|---|---|---|
| Stickers | $6.75 | 45 | 3 | $0.45 |
| Card | $4.80 | 60 sheets | 1 | |
| Ribbon | $3.60 | 300 cm | 12 cm | |
| Ink | $9.00 | 150 prints | 2 | |
| Bag | $5.40 | 90 bags | 1 | |
| Label | $2.10 | 70 labels | 1 | |
| **TOTAL per item** | | | | |

## The ones that go wrong

About one in every ten comes out badly and cannot be sold. You still paid for it.

| Question | Answer |
|---|---|
| Cost of materials for 10 items | |
| How many are actually sellable | |
| Real cost of one sellable item | |

That real cost is the one to use. Businesses call the waste **spoilage** and they all plan for it.

## Which is bigger?

Your cost per item before spoilage, and after. How much did it move, in cents?

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### TALK ABOUT IT

Ask an adult about something at their work that gets thrown away or wasted.

Ask whether anyone counts it.

### STUCK? TRY THIS

Divide first, multiply second. Pack price divided by how many in the pack gives you one. Then multiply by how many you use.

### ★ CHALLENGE ZONE

A supplier offers a bigger pack: 200 stickers for $24.00 instead of 45 for $6.75.

Work out the cost per sticker both ways. Then say what you would need to know before buying the big pack. There is more than one right answer.

WEEK 4 MONEY

# Margin

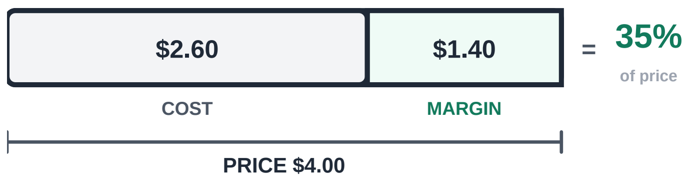
Not just the gap. The gap as a share of the price, which is the number people compare.

**IN CLASS THIS WEEK**
Multiplying fractions and finding a fraction of a quantity (5.NF.4, 5.NF.6)

**MONEY WARM-UP · 2 MINUTES**
Price $4.00, cost $2.60. What is the gap? What fraction of the price is it? ________

**WORDS TO KEEP**

| | |
|---|---|
| **margin** | Price minus cost, and the share of the price that leaves. |
| **share of price** | Out of every dollar a customer hands you, how much stays. |




## Two ways to say the same sale

The gap is **$1.40**. As a share of the $4.00 price it is **about a third**, which businesses write as 35%.

Why bother with the share? Because it lets you compare two things that cost completely different amounts. A $1.40 gap is excellent on a $4 item and terrible on a $40 one.

## Fill the table

| Item | Price | Cost | Margin | Share of price |
|---|---|---|---|---|
| Card | $4.00 | $2.60 | $1.40 | about 1/3 |
| Bracelet | $3.00 | $1.00 | | |
| Bag | $5.00 | $4.00 | | |
| Badge | $2.00 | $0.50 | | |

## Which is the better product?

Look at your table. Rank the four by margin in dollars, then rank them again by share of price.

| Rank | By dollars | By share |
|---|---|---|
| 1 (best) | | |
| 2 | | |
| 3 | | |
| 4 | | |

## They disagree

The two rankings are not the same. Explain why, and say which one you would trust when choosing what to make.

______________________________

______________________________

### 💬 TALK ABOUT IT

Ask an adult to guess what share of a restaurant meal's price is the food itself.

Ask what the rest of the price has to pay for.

### STUCK? TRY THIS

Share of price means: out of every dollar the customer hands you, how much stays? Work out the gap first, then compare it to the price.

### ★ CHALLENGE ZONE

Two products. One has a $2.00 margin and sells 10. One has a $0.80 margin and sells 40.

Work out the total margin from each. Then say which product you would rather have, and what extra fact would change your mind.

WEEK 5 MONEY

# Break Even

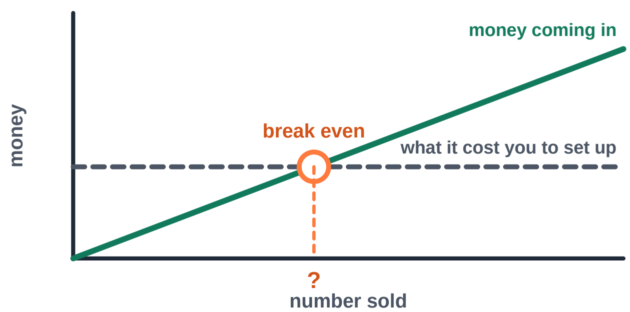
How many you must sell before you have made a single cent.

**IN CLASS THIS WEEK**
Dividing to solve problems (5.NBT.6, 5.OA.3)

**MONEY WARM-UP · 2 MINUTES**
Fixed costs are $24.00. You make $1.50 margin on each sale. How many sales before you have covered the $24? ________

**WORDS TO KEEP**

| | |
|---|---|
| **break even** | The number of sales that covers your fixed costs exactly. |
| **loss** | Anything below break even. A real result, not a disgrace. |




## The most useful number in the book

Your fixed costs are already spent before you sell anything. Every sale after that pays some of them back.

**FIXED COSTS ÷ MARGIN ON ONE = BREAK EVEN**

Below that number you have lost money. Above it you are making some. It is the first thing an adult asks about any small business.

## Work these out

| Fixed costs | Margin on one | Break even at |
|---|---|---|
| $24.00 | $1.50 | 16 |
| $18.00 | $0.90 | |
| $30.00 | $2.50 | |
| $12.50 | $0.50 | |

## When it does not divide evenly

Fixed costs $20.00, margin $1.50. Work out $20.00 ÷ $1.50.

You cannot sell part of an item, so round **up**. Why up and not down?

## Is it reachable?

Your break even is 34 items and about 60 people will walk past your table. Is that plan sensible? Say what you would change.

**TALK ABOUT IT**

Ask an adult whether anything they have organized had to cover a cost before it made anything: a school trip, a raffle, a club.

Ask what happened when not enough people signed up.

**STUCK? TRY THIS**

Fixed costs divided by the margin on one sale. Not divided by the price. It is the margin that pays the fixed costs back.

**★ CHALLENGE ZONE**

You could cut your sign from $9.00 to $3.00 by making it yourself out of paper.

Work out how much your break-even number falls. Then say what a worse sign might cost you in sales, and whether the trade is worth it.

# The Price Ladder

Every price is a bet: fewer sales at more each, or more sales at less each.

**IN CLASS THIS WEEK**
Multiplying decimals; patterns (5.NBT.7, 5.OA.3)

**MONEY WARM-UP · 2 MINUTES**
At $3.00 you sell 30. At $4.00 you sell 22. Which brings in more money? ________

**WORDS TO KEEP**

**price ladder** Every price you could charge, with what would sell at each.

### Raising the price does two things

It gives you more from every sale, and it gives you fewer sales. The only way to know which wins is to work it out.

Cost is $1.20 an item. Fill in the ladder and find the best row. It is usually not the top one and it is usually not the bottom one.

### The ladder

| Price | You would sell | Money in | Cost of those | Profit |
|---|---|---|---|---|
| $1.50 | 46 | | | |
| $2.00 | 40 | | | |
| $2.50 | 35 | | | |
| $3.00 | 30 | | | |
| $4.00 | 22 | | | |
| $5.00 | 12 | | | |
| $6.00 | 5 | | | |

## Read your own table

| Question | Answer |
|---|---|
| Which price brings in the most money? | |
| Which price makes the most profit? | |
| Are they the same row? | |
| Which price would you choose? | |

## The two numbers people confuse

Money in and profit peak at different prices in this table. Explain to somebody why that happens.

___

___

### 💬 TALK ABOUT IT

Ask an adult about something they stopped buying because the price went up.

Ask about something whose price went up that they kept buying anyway. What is the difference between the two things?

### STUCK? TRY THIS

Do one row completely before starting the next. Money in is price times how many. Cost is $1.20 times how many. Profit is the first minus the second.

### ★ CHALLENGE ZONE

Your table only guesses how many would sell at each price.

Write down two ways you could find out for real instead of guessing, and say which one you could actually do before market day.

WEEK 7 MEASURING

# Reading a Sales Chart

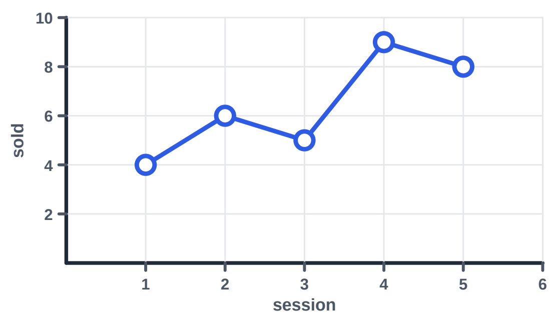
Numbers in a row hide the shape. Plotted on a grid, the shape is obvious.

**IN CLASS THIS WEEK**
Plotting ordered pairs on the coordinate plane (5.G.1, 5.G.2)

**MONEY WARM-UP · 2 MINUTES**
A stall sold 4, 6, 5, 9 and 8 items in five sessions. What is the total? ________

**WORDS TO KEEP**

| | |
|---|---|
| **ordered pair** | Across first, then up. Session 4 with 9 sold is (4, 9). |
| **trend** | Which way the points are going, which a total cannot show you. |




### Every point is a pair

Session 4, nine sold. That is the ordered pair **(4, 9)**: across first, then up.

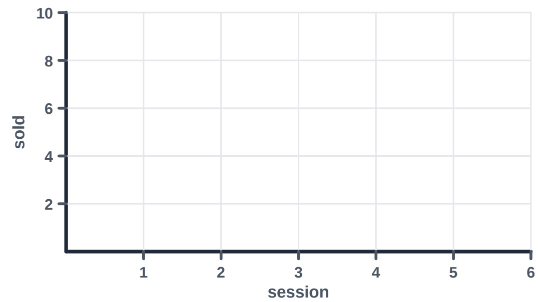
Once the points are on the grid you can see things a list of numbers hides. Where it dipped. Whether it is climbing. Which session was unlike the rest.

### Read the chart above

| Question | Answer |
|---|---|
| Which session sold the most? | |
| Which sold the fewest? | |
| Between which two sessions did it rise the most? | |
| Write session 2 as an ordered pair | |

## Plot your own

A second stall sold 2, 5, 7, 8 and 10. Plot it, then join the points.




## Compare the two

Which stall would you rather run, and why? The totals are close, so answer using the **shape**.

___

___

💬 **TALK ABOUT IT**

Ask an adult if anything at their work is tracked on a chart.

Ask what they look for when they see it: the total, or which way it is going?

**STUCK? TRY THIS**

Across first, then up. The first number in the pair is how far along the bottom, the second is how far up.

★ **CHALLENGE ZONE**

The second stall sold fewer in total but the shape is different.

Say which stall you would bet on for session 6, and explain your reasoning using the chart rather than the total.

WEEK 8 MEASURING

# The Average Sale

One number that stands in for a whole day, and what it hides.

**IN CLASS THIS WEEK**
Dividing to find an average (5.NBT.6)

**MONEY WARM-UP · 2 MINUTES**
Six customers spent $3.00, $4.50, $2.00, $7.00, $3.50 and $4.00. What did they spend altogether? ________

**WORDS TO KEEP**

**average** Total divided by how many. What each would be if all were equal.

## Add them up, share them out

The **average** sale is the total money divided by the number of customers. It answers: if every customer had spent the same, what would it have been?

It is useful because it lets you plan. If your average sale is $4.00 and you need $80, you need about 20 customers.

## Work it out

| Customer | Spent |
|---|---|
| 1 | $3.00 |
| 2 | $4.50 |
| 3 | $2.00 |
| 4 | $7.00 |
| 5 | $3.50 |
| 6 | $4.00 |
| **TOTAL** | |

| Question | Answer |
|---|---|
| Number of customers | |
| **Average sale** | |

Now plan with it. To take $60 at that average, how many customers do you need? ________

## What the average hides

Two stalls both averaged $4.00 a customer.

| Stall | What the six customers spent |
|---|---|
| A | $4, $4, $4, $4, $4, $4 |
| B | $1, $1, $1, $2, $3, $16 |

Same average, completely different days. Which stall would you rather run next week, and why?

## Two ways to raise it

Name one way to raise your average sale that is not raising your price.

**💬 TALK ABOUT IT**

Ask an adult where they have heard an average used to describe a lot of people.

Ask whether they thought it described them.

**STUCK? TRY THIS**

Total first, then divide by how many. If the division does not come out exactly, give your answer to the nearest cent.

**★ CHALLENGE ZONE**

Stall B's day depended almost entirely on one customer spending $16.

Work out stall B's average **without** that customer. Then say what that tells you about how much to trust an average from a small number of sales.

WEEK 9 MONEY

# Two Products, One Table

The moment you sell more than one thing, you need to know which one is carrying you.

**IN CLASS THIS WEEK**
Operations with decimals (5.NBT.7)

**MONEY WARM-UP · 2 MINUTES**
You sold 14 cards at $2.50 and 9 badges at $1.50. How much came in altogether? ________

**WORDS TO KEEP**

**product line** More than one thing sold to the same customer.

**total margin** Margin on one, times how many sold. The number that pays you.

## A line, not a product

Last year you sold one thing. This year you will sell three, and they will not perform the same.

One will sell in numbers and make almost nothing each. One will hardly sell but pay well when it does. Only a table tells you which is which, and the answer surprises people every time.

## Fill it in

| Product | Sold | Price | Cost | Money in | Margin each | Total margin |
|---|---|---|---|---|---|---|
| Cards | 14 | $2.50 | $1.10 | | | |
| Badges | 9 | $1.50 | $0.40 | | | |
| Bags | 4 | $6.00 | $3.80 | | | |
| Keyrings | 21 | $1.00 | $0.55 | | | |
| **ALL** | | | | | | |

## Now answer the real questions

| Question | Answer |
| --- | --- |
| Which sold the most? | |
| Which brought in the most money? | |
| Which made the most total margin? | |
| Which made the most margin per sale? | |

Four questions, and the answer is not the same product every time. That is the point of the table.

## Drop one

If you had to stop making one of the three, which would it be? Give a reason from the table, not a feeling.

________________________________________

________________________________________

**▶ ADD TO YOUR NUMBERS FILE**

Copy FIXED COSTS ÷ MARGIN = BREAK EVEN onto the inside cover of your Numbers File. You will use it in nine more weeks this year.

**💬 TALK ABOUT IT**

Show an adult your finished table and ask which product they would drop.

See whether they picked the same one you did, and why.

**STUCK? TRY THIS**

Do one row all the way across before starting the next. Money in is sold times price. Margin each is price minus cost. Total margin is margin each times sold.

**★ CHALLENGE ZONE**

Bags sell in tiny numbers but make the most margin per sale.

Work out how many extra bags you would have to sell to beat the cards on total margin. Then say whether that is realistic, using the numbers in the table.

WEEK 10 MEASURING

# Good Days and Bad Days

The average tells you the middle. The spread tells you what to expect.

**IN CLASS THIS WEEK**
Line plots and interpreting data (5.MD.2)

**MONEY WARM-UP · 2 MINUTES**
A stall sold 3, 9, 4, 11, 6, 8 and 5 over seven days. What is the biggest number minus the smallest?

____

**WORDS TO KEEP**

**range** Biggest minus smallest. What the average cannot tell you.

### Two stalls, same average

Both of these averaged 6 sales a day over a week. They are not the same business.

| Stall | Mon | Tue | Wed | Thu | Fri | Sat | Sun |
|---|---|---|---|---|---|---|---|
| A | 6 | 5 | 7 | 6 | 6 | 7 | 5 |
| B | 1 | 12 | 2 | 11 | 1 | 13 | 2 |

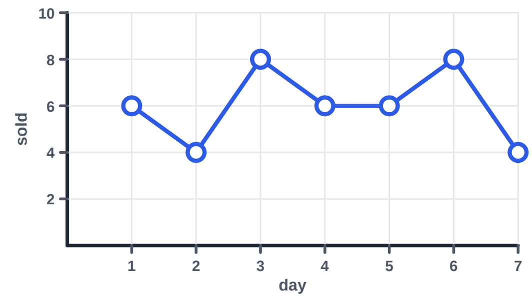
The **range** is the biggest minus the smallest. Work it out for each.

| | Range |
|---|---|
| Stall A | |
| Stall B | |

### Why it matters for stock

You have to decide how many to bring before you know how many will sell. Answer for each stall.

| Question | Stall A | Stall B |
|---|---|---|
| Fewest I might sell | | |
| Most I might sell | | |
| How many would you bring? | | |




## The cost of guessing wrong

| If you bring | and sell | what happens? |
| --- | --- | --- |
| Too many | fewer | |
| Too few | more | |

## Which is worse?

Bringing too many, or too few? Your answer should depend on what you are selling. Say what would change it.

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💬 **TALK ABOUT IT**

Ask an adult about a place that sells out early, and a place that always has plenty.

Ask which one annoys them more, and why.

**STUCK? TRY THIS**

Range is one subtraction: largest minus smallest. It does not need the average at all.

★ **CHALLENGE ZONE**

Stall B sold the same total as stall A but half its days were nearly nothing.

Write down one kind of business that really behaves like stall B, and say what its owner has to do differently because of it.

WEEK 11 MEASURING

# Who Else Is Selling?

Your customers have a fixed amount to spend and other stalls want it too.

**IN CLASS THIS WEEK**
Fractions of a whole; comparing fractions (5.NF.1, 5.NF.2)

**MONEY WARM-UP · 2 MINUTES**
60 shoppers each have $8 to spend. How much money is in the room altogether? ________

**WORDS TO KEEP**

**share** Your slice of all the money in the room.

**even share** Total money divided by number of stalls. A ruler, not a target.

### The money in the room is a number

At your market every shopper is given the same amount, so the total money is something you can actually work out. It does not grow because you want it to.

Which means every stall is competing for slices of the same amount. Your **share** is the slice you get.


### Work out the room

| Question | Answer |
|---|---|
| Shoppers | 60 |
| Each is given | $8.00 |
| Total money in the room | |
| Number of stalls | 12 |
| An even share for each stall | |

## Above or below an even share?

Your plan is to take $60. Compare that with an even share.

| Question | Answer |
|---|---|
| An even share is | |
| My plan is | $60.00 |
| That is what fraction of the room? | |
| Above or below even? | |

## What would justify it?

Planning above an even share is allowed, but it needs a reason. Write yours, or lower your plan.

______________________________________________

______________________________________________

**TALK ABOUT IT**

Ask an adult about two shops near each other that sell similar things.

Ask how each one is different, and which they go to.

**STUCK? TRY THIS**

An even share is the total money divided by the number of stalls. It is not a target. It is a ruler for checking whether your plan is sane.

**★ CHALLENGE ZONE**

Twelve stalls, and four of them are selling roughly the same thing as you.

Work out the even share among just those five. Then say what you could change so that a shopper picks yours, and note that lowering the price is only one of the answers.

WEEK 12 REVIEW

# The Numbers Detective

Everything from weeks 1 to 11, used on one real business you can walk to.

**IN CLASS THIS WEEK**
Review: decimals, division, margin, break even (5.NBT.6, 5.NBT.7, 5.NF.4)

**MONEY WARM-UP · 2 MINUTES**
A stall buys 80 items at $1.35 each and sells them all at $3.00. What is the total margin? ________

**WORDS TO KEEP**

**estimate** A number you can explain. Worth far more than a blank.

## Your assignment

Pick a real small business you can see. Estimate its numbers the way an analyst would, and mark every guess with a **?** so you remember which figures you actually know.

## The file

| What you are working out | Your estimate |
|---|---|
| What it sells (name two or three things) | |
| Roughly what one of them costs to make or buy | |
| What it charges | |
| Margin on one, in dollars | |
| Margin as a share of the price | |
| Fixed costs it pays whether or not anyone comes | |
| Roughly how many it needs to sell to break even | |

### The two questions that matter

| Question | Your answer and reasoning |
|---|---|
| Which of its products do you think pays best? | |
| What would you change first? | |

### How confident are you?

Count how many rows in your file are guesses. Out of seven, how many did you actually know? ________

Name the one fact that, if you could ask the owner, would improve your estimate most.

______________________________

**▶ ADD TO YOUR NUMBERS FILE**

Put your detective file in your Numbers File. It is the last week you spend on somebody else's business. Next week you start your own, with a partner.

**💬 TALK ABOUT IT**

Show your file to an adult and ask which of your estimates looks wrong.

Ask them what the business pays for that you did not put on the list.

**STUCK? TRY THIS**

You will not be able to find the real numbers, and that is fine. An estimate you can explain beats a blank every time. Mark it with a question mark and move on.

**★ CHALLENGE ZONE**

Add rent, electricity and the owner's own time to your fixed costs.

Redo the break-even number with those included. Most small businesses look very different once you do. Write one sentence about what you found.

WEEK 13 BUILDING

# Your Team, Your Roles

Trimester 2 starts here. Two or three of you, one business, and jobs decided now.

**IN CLASS THIS WEEK**
Adding and subtracting fractions with unlike denominators (5.NF.1)

**MONEY WARM-UP · 2 MINUTES**
Three partners split a profit. One takes 1/2, one takes 1/4. What fraction is left for the third?

______

**WORDS TO KEEP**

**partner** Someone who shares the work, the decisions and the profit.

## Decide it now, not in June

Every argument a small partnership has is about one of three things: who does what, who decides, and who gets what.

All three are easy to agree in September and impossible to agree in June with money on the table. So you write them down this week and sign it.

## Who does what

| Job | Whose job | Backup |
|---|---|---|
| Making the products | | |
| Keeping the numbers | | |
| The sign and the table | | |
| Talking to customers | | |
| The money box | | |
| Buying the materials | | |
| Packing and carrying it | | |

## Splitting the profit

Write your split as fractions. They must add to one whole, and you have to show that they do.

partner 1 partner 2 partner 3

| Partner | Fraction | Why that share |
|---|---|---|
| | | |
| | | |
| | | |
| **Adds to** | | |

## Two hard questions

Who decides if you disagree? And what happens if somebody stops doing their job?

---

---

**▶ ADD TO YOUR NUMBERS FILE**

Page 1 of your Numbers File: your roles, your profit split written as fractions that add to one, and both answers to the hard questions. Sign it.

**💬 TALK ABOUT IT**

Ask an adult who has worked in a team what made it work, or not work.

Ask whether it was about the work or about who decided.

**STUCK? TRY THIS**

An equal split is the easiest to agree and the easiest to defend. If you want an unequal one, the reason has to be about work done, not about whose idea it was.

**★ CHALLENGE ZONE**

One partner does most of the making. Another finds most of the customers.

Argue for an equal split in two sentences, then argue against it in two. Then decide, and say which argument you found harder to make.

WEEK 14 BUILDING

# The Problem You Will Solve

One problem, one customer, and a customer you can reach more than once.

**IN CLASS THIS WEEK**
Multi-step problems with decimals (5.NBT.7)

**MONEY WARM-UP · 2 MINUTES**
You have $30.00 to start. Materials cost $1.85 per item. How many can you make? ________

**WORDS TO KEEP**
**repeat customer** Somebody you can sell to twice. This year that is the rule.

## The new rule this year

Your customer has to be somebody you can sell to **twice**. This year you run two selling sessions, not one, so a one-off crowd will not do.

That rules out some good ideas and it is meant to. Repeat customers are where the interesting arithmetic lives.

## Three candidates

| The problem | Who has it | Can I reach them twice? | Score /3 |
|---|---|---|---|
| 1. | | | |
| 2. | | | |
| 3. | | | |

## The one you are committing to

Write it as one sentence naming who, what goes wrong, and why it is worth fixing.

______________________________

______________________________

## What they do about it now

Nobody has an unsolved problem. They have a bad solution. Find out what it is by asking three people.

| Who I asked | What they do about it now |
|---|---|
| 1. | |
| 2. | |
| 3. | |

## Your opening

What is wrong with what they do now? That gap is the only reason anyone would switch to you.

________________________________________

**▶ ADD TO YOUR NUMBERS FILE**

Page 2 of your Numbers File: your problem in one sentence, your customer, and what three real people told you they do about it now.

**💬 TALK ABOUT IT**

Explain your problem to an adult in under thirty seconds.

Ask whether they have ever had it. Write down what they say even if it is no.

**STUCK? TRY THIS**

If you cannot name a specific place and time where you would find your customer a second time, the problem is not ready. Make it smaller and more local.

**★ CHALLENGE ZONE**

Your problem may already have a solution people are happy with.

Find out what it costs. If you cannot beat it on price, write down the two other things you could beat it on.

WEEK 15 BUILDING

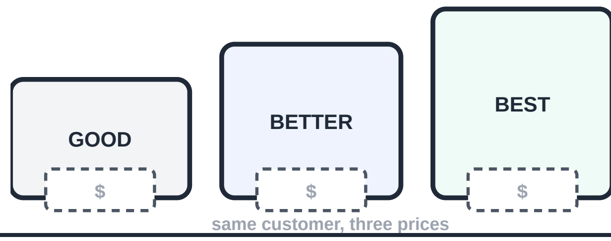
# Good, Better, Best

Three products, one customer. Not three businesses.

**IN CLASS THIS WEEK**
Multiplying fractions and decimals (5.NF.4, 5.NBT.7)

**MONEY WARM-UP · 2 MINUTES**
A small costs $1.20, a large costs 1 and 1/2 times that. What does the large cost? ________

**WORDS TO KEEP**
**good, better, best** Three versions at three prices, for one customer.




## Why three

A customer who says no to $5.00 may say yes to $2.00. A customer who was always going to buy might have paid $8.00.

One price catches one kind of customer. Three prices catch three. The trick is that all three must be for the **same person** solving the **same problem**, or you have started three businesses by accident.

## Your three

| | What it is | Roughly what price? |
|---|---|---|
| GOOD | | |
| BETTER | | |
| BEST | | |

## The test

| Question | Yes or no | If no, fix it |
|---|---|---|
| Same customer for all three? | | |
| Same problem for all three? | | |
| Can I actually make all three? | | |
| Is BEST clearly better than GOOD? | | |

## What makes BEST better?

Bigger, or more of them, or nicer, or faster? Name the one thing, because the customer has to see it in a second.

---

**▶ ADD TO YOUR NUMBERS FILE**

Page 3 of your Numbers File: your three products, what each one is, and what makes BEST better than GOOD in one word.

**💬 TALK ABOUT IT**

Ask an adult to find something at home that comes in three sizes or versions.

Ask which one they buy, and why not one of the others.

**STUCK? TRY THIS**

Start from the middle one. Design BETTER first, then make a cheaper version and a nicer version of the same thing.

**★ CHALLENGE ZONE**

Shops often make the middle option the one they most want you to buy.

Write down how the other two could be designed to make the middle one look like the obvious choice. This is a real technique, and noticing it is the point.

# Build All Three Rough

Three bad versions this week beats one good one.

**IN CLASS THIS WEEK**
Converting measurement units (5.MD.1)

**MONEY WARM-UP · 2 MINUTES**
A ribbon is 3 meters long. Each item needs 25 cm. How many items from one ribbon? ________

**WORDS TO KEEP**

**prototype** A rough version made fast, so you find out what is hard to build.

## Rough on purpose, three times

Make all three now, badly and fast. You are not trying to impress anyone; you are trying to find out which one is hard to make.

That matters more than it sounds. A product that takes twice as long is a product you will bring half as many of.

## Time yourself making one of each

| | Minutes to make one | What went wrong | Materials used |
|---|---|---|---|
| GOOD | | | |
| BETTER | | | |
| BEST | | | |

## Measure them

Write every measurement in the same unit before you compare. Centimeters are usually easiest.

| | Size | In cm |
|---|---|---|
| GOOD | | |
| BETTER | | |
| BEST | | |

## How many could you make in an hour?

| | Minutes each | In 60 minutes |
|---|---|---|
| GOOD | | |
| BETTER | | |
| BEST | | |

Divide 60 by your minutes each. Round down: half an item is not a thing you can sell.

## The surprise

Which one took much longer than you expected? What would you change to speed it up?

---

---

**▶ ADD TO YOUR NUMBERS FILE**

Page 4 of your Numbers File: your three rough versions, timed and measured, with a note on what went wrong building each.

**💬 TALK ABOUT IT**

Show all three to somebody at home without saying which is which.

Ask them to put them in order of price. See whether they get the order you intended.

**STUCK? TRY THIS**

Time it with a clock, not a guess. Nearly everyone underestimates by about half, and the numbers in weeks 22 and 23 are built on this measurement.

**★ CHALLENGE ZONE**

BEST takes three times as long as GOOD to make.

Work out whether you would make more money spending an hour on BEST or an hour on GOOD, using your rough prices from week 15. Show the arithmetic both ways.

WEEK 17 BUILDING

# Score It With Real People

Eight people, three products, one scoring sheet. Numbers, not opinions.

**IN CLASS THIS WEEK**
Collecting and interpreting data (5.MD.2)

**MONEY WARM-UP · 2 MINUTES**
Eight people score a product out of 5. The scores total 28. What is the average? ________

**WORDS TO KEEP**

**signal** Something two or more people said. One person saying it is an opinion.

## Show, do not explain

Put all three in front of somebody and say nothing except the price. Everything you have to explain is something a customer will not understand either.

Then ask two questions and write the answers down exactly.

## The scoring sheet

Score out of 5: would you buy this? Ask eight people.

| Person | GOOD | BETTER | BEST | Which would you actually buy? |
|---|---|---|---|---|
| 1 | | | | |
| 2 | | | | |
| 3 | | | | |
| 4 | | | | |
| 5 | | | | |
| 6 | | | | |
| 7 | | | | |
| 8 | | | | |
| **TOTAL** | | | | |

## Work out the averages

| | Total score | Average out of 5 | Votes to buy |
|---|---|---|---|
| GOOD | | | |
| BETTER | | | |
| BEST | | | |

The highest average and the most votes to buy are often different products. Which was it for you?

______________________________

## What two or more people said

Anything said twice is a signal. Anything said once is an opinion. Write the signals.

______________________________

______________________________

**▶ ADD TO YOUR NUMBERS FILE**

Page 5 of your Numbers File: all eight scoring rows, the three averages, and the signals you heard more than once.

**💬 TALK ABOUT IT**

Ask an adult about a time they liked something but did not buy it.

Ask what the difference was between liking it and paying for it.

**STUCK? TRY THIS**

A high score and an actual purchase are different things. People are generous with scores and careful with money, so the final column matters most.

**★ CHALLENGE ZONE**

One product scored highest but almost nobody chose to buy it.

Write down two reasons that could happen. One should be about price, and one should not be.

WEEK 18 BUILDING

# Cut One

Three is too many to make well. Kill the weakest one yourself, on the numbers.

**IN CLASS THIS WEEK**
Comparing decimals and fractions (5.NBT.3, 5.NF.2)

**MONEY WARM-UP · 2 MINUTES**
Product A scores 3.5 out of 5. Product B scores 7/2 out of 5. Which is higher? ________

**WORDS TO KEEP**

**cutting** Stopping the weakest idea on purpose, before it costs you.

## Cutting is a skill

Most beginners keep everything because dropping an idea feels like failing. Keeping a weak product costs you the time you needed for the strong ones.

So this week you cut one on purpose, using the week 16 and 17 numbers rather than which one you liked making.

## Put the evidence in one place

Copy these across from weeks 16 and 17. Do not go from memory.

| | Avg score | Votes to buy | Minutes to make | Rough margin |
|---|---|---|---|---|
| GOOD | | | | |
| BETTER | | | | |
| BEST | | | | |

Now work out the one number that puts them on the same footing: margin per minute of your time.

| | Rough margin | Minutes to make | Margin per minute |
|---|---|---|---|
| GOOD | | | |
| BETTER | | | |
| BEST | | | |

## The decision

Cut one, and keep two. Write which, and give one reason from each of two different columns.

| | Your answer |
|---|---|
| I am cutting | |
| Reason from the scores | |
| Reason from the time or the margin | |

## What you gain

Work out how many extra of your remaining products you could make with the time the cut one would have taken.

▶ **ADD TO YOUR NUMBERS FILE**

Page 6 of your Numbers File: the comparison table, which product you cut, and the two reasons.

**TALK ABOUT IT**

Tell an adult which one you cut and why.

Ask them about something they gave up on, and whether they wish they had done it sooner or later.

**STUCK? TRY THIS**

If two products score the same, cut the one that takes longer to make. Time is the material you cannot buy more of before market day.

★ **CHALLENGE ZONE**

Somebody in your team wants to keep the product you are cutting.

Write down the strongest argument for keeping it, then say what evidence would change your mind. If no evidence would, you are not deciding on the numbers.

WEEK 19 MONEY

# Your All-In Cost

Every material, divided down, with the wasted ones paid for. To the penny.

**IN CLASS THIS WEEK**
All four operations with decimals (5.NBT.7)

**MONEY WARM-UP · 2 MINUTES**
A pack of 24 costs $8.40. You use 2 per item. What is the cost per item? ________

**WORDS TO KEEP**

**all-in cost** Materials, packaging and waste, for one sellable item.

**◂ YOU WILL NEED, FROM YOUR NUMBERS FILE**
- page 3: your three products, and the one you cut
- page 4: what each one is made of and how long it takes

## Do this for both products

One table each. Pack price divided by how many in the pack, times how many you use. Add packaging. Then add the waste.

| Material | Pack price | How many in a pack | Used per item | Cost per item |
|---|---|---|---|---|
| Example | $8.40 | 24 | 2 | $0.70 |
| | | | | |
| | | | | |
| | | | | |
| | | | | |
| **Materials subtotal** | | | | |

## The two lines people forget

| | Product 1 | Product 2 |
|---|---|---|
| Materials subtotal | | |
| Packaging: bag, label, tie | | |
| Waste: about 1 in 10 goes wrong | | |
| **ALL-IN COST OF ONE** | | |

### How the waste line is worked out

If you make 10 and one is unsellable, you paid for 10 and can sell 9. Divide the cost of 10 by 9.

| Question | Product 1 | Product 2 |
|---|---|---|
| Materials and packaging for 10 | | |
| How many are sellable | | |
| Real cost of one sellable item | | |

### Compare with your guess

In week 15 you guessed roughly what these would cost. How far out were you, in cents?

---

**▶ ADD TO YOUR NUMBERS FILE**

Page 7 of your Numbers File: two full materials tables and the all-in cost of one of each, to the penny.

**💬 TALK ABOUT IT**

Show an adult your materials table and ask what you left off.

Adults are very good at this. Write down whatever they spot, even if you disagree.

**STUCK? TRY THIS**

Divide before you multiply. Pack price divided by how many are in the pack gives the cost of one, and only then multiply by how many you use.

**★ CHALLENGE ZONE**

Your waste rate is a guess. Suppose it is really 1 in 5, not 1 in 10.

Redo the all-in cost of one for both products at that rate. Then say which product is hurt more by waste, and why.

WEEK 20 MONEY

# What It Costs Before You Sell Anything

Add up everything you have to pay for whether or not one person turns up.

**IN CLASS THIS WEEK**
Adding decimals; multi-step problems (5.NBT.7)

**MONEY WARM-UP · 2 MINUTES**
Sign $6.50, labels $3.20, table cloth $4.00. What are the fixed costs altogether? ________

**WORDS TO KEEP**

**fixed costs** Everything you pay before a single customer turns up.

**◂ YOU WILL NEED, FROM YOUR NUMBERS FILE**

- page 7: the all-in cost of one of each


## Your fixed costs

These are already spent on the morning of your first session. Every one of them has to be earned back before you make a cent.

| Fixed cost | Amount |
|---|---|
| Sign and any decoration | |
| Price labels, printed once | |
| Anything for the table itself | |
| Anything you had to buy to make the first one | |
| | |
| **TOTAL FIXED COSTS** | |

## Borrowed is not free

If you borrow something and have to give it back or replace it, write down what it would cost to replace. Which of yours is borrowed?

______________________________

## What the fixed cost does to one item

Spread your fixed costs over different numbers of sales.

| If you sell | Fixed cost per item | Plus all-in cost | Real cost of one |
|---|---|---|---|
| 10 | | | |
| 25 | | | |
| 50 | | | |

## The trap

Your price has to beat the real cost of one, not just the all-in cost. At which of those three sales numbers does your week 15 price still work?

**▶ ADD TO YOUR NUMBERS FILE**

Page 8 of your Numbers File: your full fixed cost list with a total, and the table showing what fixed costs do to the real cost of one item.

**💬 TALK ABOUT IT**

Ask an adult what their workplace pays for on a day when no customers come at all.

Ask whether those costs stop when things are quiet.

**STUCK? TRY THIS**

A cost is fixed if you would still pay it having sold nothing. Test each one with that question and the list sorts itself.

**★ CHALLENGE ZONE**

You could spend $12 on a really good sign or $3 on a plain one.

Work out how many extra sales the good sign would have to bring in to be worth it, using your own margin. Then say whether you believe it would.

WEEK 21 MONEY

# Pricing the Line

Two products, two prices, and they have to make sense next to each other.

**IN CLASS THIS WEEK**
Operations with decimals; comparing (5.NBT.7, 5.NBT.3)

**MONEY WARM-UP · 2 MINUTES**
Cost $1.35. You want a margin of about half the price. What price would do that? ________

**WORDS TO KEEP**

**price point** One chosen price. A line has several, on purpose.

**◂ YOU WILL NEED, FROM YOUR NUMBERS FILE**

- page 5: what your eight testers scored and said
- page 7: the all-in cost of one of each


## Three ways in, then a decision

| Method | What it gives you | Product 1 | Product 2 |
|---|---|---|---|
| Cost plus | all-in cost plus what you want to keep | | |
| Look around | what similar things sell for | | |
| Ask them | what your week 17 people said | | |

## Your prices

| | All-in cost | Price | Margin | Share of price |
|---|---|---|---|---|
| Product 1 | | | | |
| Product 2 | | | | |

## Do they work as a pair?

| Question | Yes or no | If no, what changes? |
| --- | --- | --- |
| Is the better one clearly more expensive? | | |
| Is the gap between them big enough to notice? | | |
| Is each price above its real cost? | | |
| Are both easy to pay in coins? | | |

## The cheap one has a job

The lower-priced product is often there to get somebody to stop and buy anything at all. Does yours do that? Say how.

______________________________

▶ **ADD TO YOUR NUMBERS FILE**

Page 9 of your Numbers File: both prices, both margins, the share of price for each, and one sentence on why those two prices work together.

💬 **TALK ABOUT IT**

Ask an adult about two versions of something where they chose the more expensive one.

Ask what made the extra money feel worth it.

**STUCK? TRY THIS**

Price the better product first, then set the cheaper one clearly below it. Doing it the other way round usually leaves the two prices too close together.

★ **CHALLENGE ZONE**

If the two prices are very close, most people buy the better one and you sell almost no cheap ones.

Work out whether that would actually be bad for you, using your own margins. The answer is not obvious, so show the arithmetic.

WEEK 22 MONEY

# Your Break-Even, In Units

The number that decides whether your plan is sensible or a wish.

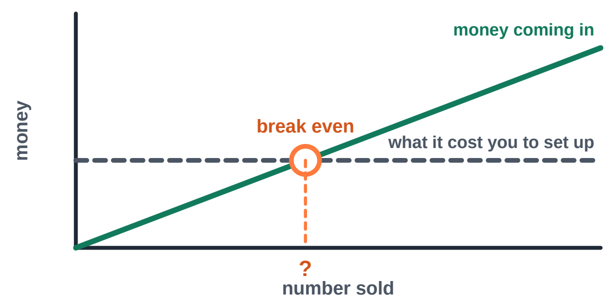
**IN CLASS THIS WEEK**
Dividing decimals; rounding up in context (5.NBT.7)

**MONEY WARM-UP · 2 MINUTES**
Fixed costs $21.00, margin $1.75 each. How many sales to break even? ________

**WORDS TO KEEP**
**break-even units** How many of each you must sell before there is any profit.

**◂ YOU WILL NEED, FROM YOUR NUMBERS FILE**

- page 8: your total fixed costs
- page 9: your two prices and both margins




## Your own number

| | Your figure |
|---|---|
| Total fixed costs (week 20) | |
| Margin on product 1 (week 21) | |
| Margin on product 2 (week 21) | |
| Average margin across the two | |
| **BREAK EVEN, IN ITEMS** | |

## Round up, always

If it comes out at 13.4, your break even is 14. Say why you cannot round down.

______________________________

## Is it reachable?

| Question | Your answer |
| --- | --- |
| How many people will walk past your table? | |
| What fraction of them would need to buy? | |
| Is that realistic? Yes or no. | |
| If no, what are you changing? | |

## Two ways to lower it

Break even falls if fixed costs go down or margin goes up. Pick one and say exactly what you would change.

▶ **ADD TO YOUR NUMBERS FILE**

Page 10 of your Numbers File: your break-even working, the number rounded up, and your honest answer about whether it is reachable.

💬 **TALK ABOUT IT**

Tell an adult your break-even number and how many people you expect.

Ask whether they would put their own money into your plan, and why.

**STUCK? TRY THIS**

Fixed costs divided by margin, not by price. It is the margin that pays back the fixed costs, and using the price makes your break even look far better than it is.

★ **CHALLENGE ZONE**

Your break even is spread over two sessions, not one.

Work out how many you need in session one just to be halfway there. Then say what you would do if session one came in well below that.

WEEK 23 MONEY

# The Forecast

Write down what you think will happen, before it happens. Then you cannot move it.

**IN CLASS THIS WEEK**
Multiplying and adding decimals (5.NBT.7)

**MONEY WARM-UP · 2 MINUTES**
You forecast 18 at $3.00 and 7 at $5.00. What money do you forecast altogether? ________

**WORDS TO KEEP**

**forecast** What you predict will happen, written down before it does.

**◂ YOU WILL NEED, FROM YOUR NUMBERS FILE**

- page 9: your prices and margins
- page 10: your break-even number

### The most important page in this book

A guess made afterwards is not a forecast. It is a memory, and memories quietly rearrange themselves to match what happened.

So you write this now, in pen, and you do not change it. In week 31 you compare it with the truth. That comparison is where nearly all of the learning in this year actually lives.

### Session one forecast

| | How many | Price | Money in | Margin each | Total margin |
|---|---|---|---|---|---|
| Product 1 | | | | | |
| Product 2 | | | | | |
| **TOTAL** | | | | | |

## Check it against three things

| Check | Your answer | Does the forecast pass? |
|---|---|---|
| Is it above break even? | | |
| Is it below an even share of the room? | | |
| Can you actually make that many in time? | | |

If it fails any of the three, change the forecast now, not later. A forecast you already know is wrong is worth nothing.

## How sure are you?

Write your forecast total, then a lowest and a highest you would not be surprised by.

| Lowest | Forecast | Highest |
|---|---|---|
| | | |

**▶ ADD TO YOUR NUMBERS FILE**

Page 11 of your Numbers File: your session one forecast, in pen, with the date you wrote it. Do not change this page later.

**💬 TALK ABOUT IT**

Read your forecast to an adult and ask if it sounds high or low.

Ask them what they would forecast, and write their number next to yours.

**STUCK? TRY THIS**

Work out how many you can make first. A forecast above what you can physically bring is not optimism, it is an arithmetic mistake.

**★ CHALLENGE ZONE**

Most first forecasts are too high, because people forecast the day they hope for.

Write down two reasons a forecast tends to be optimistic. Then decide whether to lower yours, and say why you did or did not.

WEEK 24 REVIEW

# The Business Plan

Everything from weeks 13 to 23, on one page, with the numbers in it.

**IN CLASS THIS WEEK**
Review: decimals, fractions, division (5.NBT.7, 5.NF.1, 5.NBT.6)

**MONEY WARM-UP · 2 MINUTES**
Fixed $18.00, margin $2.25, forecast 30 sales. What profit does that forecast? ________

**WORDS TO KEEP**

**risk** Something that could go wrong. Every real plan names its three.

**◂ YOU WILL NEED, FROM YOUR NUMBERS FILE**

- every page so far. This one is the summary

## The plan

| | Your answer |
|---|---|
| Our business is called | |
| The problem we solve | |
| Who it is for | |
| Product 1, price and margin | |
| Product 2, price and margin | |
| Total fixed costs | |
| Break even, in items | |
| Forecast for session one | |
| Forecast profit if it comes true | |
| Our profit split | |

## The three risks

Name what is most likely to go wrong, and what you would do. Every real plan has this section and it is the part adults read first.

| What could go wrong | What we would do |
|---|---|
| 1. | |
| 2. | |
| 3. | |

## Read it to somebody

Read the whole page to an adult. Write down the first question they ask, because that is the weakest line on your page.

▶ **ADD TO YOUR NUMBERS FILE**

Page 12 of your Numbers File: the one-page plan. This is the page you would hand to somebody who asked what you are doing.

**TALK ABOUT IT**

Ask your reader which number on the page they trust least.

Ask what would make them trust it more. Usually the answer is evidence, not confidence.

**STUCK? TRY THIS**

Copy the numbers across from your Numbers File rather than remembering them. One wrong number copied here survives all the way to market day.

★ **CHALLENGE ZONE**

Redo the forecast profit assuming you sell only half of what you forecast.

Are you still above zero? If not, work out the smallest number of sales that keeps you there, and write it on the plan as the number that really matters.

WEEK 25 SELLING

# Your Stall and Your Story

Trimester 3. Two selling sessions, and everything from here is getting ready.

**IN CLASS THIS WEEK**

Volume and area for packing and layout (5.MD.3, 5.MD.5)

**MONEY WARM-UP · 2 MINUTES**

A box is 30 cm by 20 cm by 10 cm. What is its volume in cubic centimeters? ________

**WORDS TO KEEP**

**session** One selling stretch. This year there are two, on purpose.

**◂ YOU WILL NEED, FROM YOUR NUMBERS FILE**

- page 4: how long each product takes to make

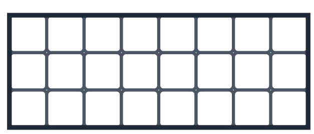
## Two sessions, not one

You sell, you measure, you change one thing, and you sell again. That is the shape of the rest of this year and it is the whole reason the forecast in week 23 exists.

It also means session one is not the end of anything. A bad first session is useful information, and you get to act on it.

## Your table, drawn to scale

Most stalls get a table about 120 cm by 60 cm. Sketch where everything goes.



each square = 15 cm

| | Yours |
|---|---|
| Table size | |
| Area of the table | |
| Space one product takes | |
| How many fit on display | |

## Packing to carry it

Work out whether your stock fits in the box you have.

| Question | Answer |
|---|---|
| Volume of my box | |
| Volume of one product | |
| How many fit, roughly | |
| Enough for the session? | |

## Your one sentence

Somebody stops and asks what you are selling. One sentence.

---

**▶ ADD TO YOUR NUMBERS FILE**

Page 13 of your Numbers File: your table plan to scale, your packing arithmetic, and your one sentence.

**💬 TALK ABOUT IT**

Set your table up at home exactly as you plan to and show an adult.

Ask what they would pick up first, and whether they can see both prices.

**STUCK? TRY THIS**

Draw the table first and put things on it second. Deciding the layout in your head and discovering it does not fit is a market-day morning problem.

**★ CHALLENGE ZONE**

You have room to display 12 items but you are bringing 40.

Say what you do with the other 28, and what the display should look like as things sell. A stall that looks empty sells more slowly than one that looks full.

WEEK 26 SELLING

# Signs and Prices People Can See

Every customer who has to ask the price is a customer you might lose.

**IN CLASS THIS WEEK**
Area; converting units (5.MD.1, 5.NF.4)

**MONEY WARM-UP · 2 MINUTES**
A sign is 45 cm by 30 cm. What is its area in square centimeters? ________

**WORDS TO KEEP**

**price display** Showing the price so nobody has to ask. Asking loses sales.

## Four things and no more

Your name. What it is. **Both prices.** One reason to care.

With two products the prices matter twice as much, because the customer is now making a choice and they will not make it if they cannot see the numbers.

## Plan the sign

| Element | What yours says | How tall are the letters? |
| --- | --- | --- |
| Business name | | |
| What you sell | | |
| Price of product 1 | | |
| Price of product 2 | | |
| One reason to care | | |

## The six-foot test

Put it on a wall and walk six feet back.

| Can you read it from six feet? | Yes or no | Fix |
|---|---|---|
| The name | | |
| What it is | | |
| The prices | | |

## Little price cards

A price card next to each product does a different job from the big sign. Say what that job is.

---

▶ **ADD TO YOUR NUMBERS FILE**

Page 14 of your Numbers File: your sign plan with letter heights and areas, and what failed the six-foot test first time.

💬 **TALK ABOUT IT**

Show your sign to somebody for three seconds, then take it away.

Ask them what it said and what the prices were. Whatever they missed needs to be bigger.

**STUCK? TRY THIS**

Prices go biggest after the name. It is the thing customers look for and the thing they are least willing to ask about.

★ **CHALLENGE ZONE**

A sheet of card is 60 cm by 45 cm and you need one big sign and four small price cards.

Work out sizes that use the sheet with the least waste. Show the areas and say how much card is left.

WEEK 27 SELLING

# The Pitch, With Numbers

Four sentences. This year one of them has a number in it.

**IN CLASS THIS WEEK**
Reporting on a topic with facts (SL.5.4)

**MONEY WARM-UP · 2 MINUTES**
You speak to 40 people in 90 minutes. Roughly how long is that each? ________

**WORDS TO KEEP**

**evidence** A real number you measured, not an adjective you chose.

## The four, plus one

Same four sentences as last year, with one addition: somewhere in there you say a real number about your own product.

the order that works

1. What is the problem?
2. What did you make?
3. Who is it for, and both prices
4. One real number about it
5. Why buy it from you?

## Write yours

| Sentence | Yours |
|---|---|
| 1. The problem | |
| 2. What we made | |
| 3. Who and how much | |
| 4. A real number | |
| 5. Why us | |

## Which number?

A real number beats an adjective. “It lasts a long time” is weaker than “it lasts about three weeks”. Pick yours from something you actually measured this year.

| Where the number could come from | Yours |
|---|---|
| How long it takes to make | |
| How long it lasts or how much it holds | |
| What the week 17 testers said | |

## Time it

Say all five out loud and time it three times. Write your best time, and what you cut to get there.

▶ **ADD TO YOUR NUMBERS FILE**

Page 15 of your Numbers File: your five sentences, the number you chose, and where that number came from.

**TALK ABOUT IT**

Give the pitch to somebody at home with your back turned so they only hear it.

Ask which sentence they would cut, and whether the number helped or sounded made up.

**STUCK? TRY THIS**

Write the five sentences down before you say them. A pitch that is different every time never gets better, because you are not practicing the same thing twice.

★ **CHALLENGE ZONE**

A number you cannot back up is worse than no number at all.

For your chosen number, write down exactly how you would answer somebody who said “how do you know that?”

WEEK 28 RUNNING IT

# Inventory and Prep

Count everything before you start, or you cannot tell what sold.

**IN CLASS THIS WEEK**
Multiplication and division with decimals (5.NBT.6, 5.NBT.7)

**MONEY WARM-UP · 2 MINUTES**
You have 34 items and expect to sell 60% of them. Roughly how many is that? ________

**◂ YOU WILL NEED, FROM YOUR NUMBERS FILE**

- page 11: your session one forecast
- page 12: the plan, for how many you said you would bring

## How the money works, again

Same as last year: the class bank lends every team the same **startup fund**. You buy materials from the class supply table with it, sell for play money and make your own change, and repay the fund at the end. What is left is yours.


## Opening inventory

Count what you are bringing. This number is half of every calculation in week 30, so count it twice.

| | Bringing | Enough for the forecast? |
|---|---|---|
| Product 1 | | |
| Product 2 | | |

## The box

Tick as you pack.

stock sign prices tally float pens

## Jobs on the day

| Job | Who | What they do if it gets busy |
| --- | --- | --- |
| Serving | | |
| Money and change | | |
| Tally sheet | | |
| Restocking the display | | |

## The float

You need change for the first customer. How much are you starting with, and in what coins?

______________________________________________

**▶ ADD TO YOUR NUMBERS FILE**

Page 16 of your Numbers File: your opening inventory counted twice, your jobs list and your float.

**💬 TALK ABOUT IT**

Walk an adult through the session minute by minute and ask what you forgot.

Ask what usually goes wrong at events like this.

**STUCK? TRY THIS**

Pack the box, then take everything out and pack it again. You find the missing thing on the second pass, not the first.

**★ CHALLENGE ZONE**

You can only display 12 items at a time but you are bringing 40.

Write the restocking rule: at how many left on the table does somebody put more out? A stall that looks empty sells more slowly than one that looks full.

WEEK 29 RUNNING IT

# Session One

Take this page to the table. Fill it in as it happens.

**IN CLASS THIS WEEK**
Recording and interpreting data (5.MD.2, 5.NBT.7)

**MONEY WARM-UP · 2 MINUTES**
Before you start, write your forecast total from week 23 here: ________

**WORDS TO KEEP**

**inventory** What you have to sell, counted before you start and after.

**traffic** How many people came past. Sales without it cannot be compared.

## Tally as it happens

One mark the moment money changes hands, in the right row. A tally filled in afterwards is a guess, and weeks 30 to 33 are all built on this page.

| | First half | Second half | Total sold |
|---|---|---|---|
| Product 1 | | | |
| Product 2 | | | |
| **BOTH** | | | |

## Count the traffic too

Every twenty minutes, roughly how many people were near your table? You need this to know whether a slow patch was you or the room.

| Time | People about | Sales in that stretch |
|---|---|---|
| First 20 min | | |
| Second 20 min | | |
| Third 20 min | | |

## What people said

| What they said, in their words | Bought? |
|---|---|
| 1. | |
| 2. | |
| 3. | |
| 4. | |
| 5. | |

## The ones who walked away

Roughly how many looked and did not buy? ________ What did most of them seem to be reacting to?

______________________________________________

▶ **ADD TO YOUR NUMBERS FILE**

Page 17 of your Numbers File: this whole page, filled in on the day.

💬 **TALK ABOUT IT**

Before you add anything up, write one sentence about how the session felt.

Then ask an adult who watched what they noticed that you did not.

**STUCK? TRY THIS**

Mark the tally as the money moves, not at the end. Everyone who has tried it the other way will tell you the same thing.

★ **CHALLENGE ZONE**

Look at your traffic column against your sales column.

Was your slowest stretch slow because fewer people came past, or because fewer of them bought? Those are completely different problems with different fixes.

WEEK 30 MONEY

# Count It Up, By Product

Not one total. One line per product, because that is where the answer is.

**IN CLASS THIS WEEK**
All four operations with decimals (5.NBT.7)

**MONEY WARM-UP · 2 MINUTES**
You sold 19 at $2.50 and 8 at $4.00. What came in altogether? ________

**WORDS TO KEEP**

**closing inventory** What came home. Opening minus closing is what you sold.

**◂ YOU WILL NEED, FROM YOUR NUMBERS FILE**

- page 16: your opening inventory
- page 17: the session one tally

## The session in one table

| | Sold | Price | Money in | All-in cost | Total cost | Margin |
|---|---|---|---|---|---|---|
| Product 1 | | | | | | |
| Product 2 | | | | | | |
| **TOTAL** | | | | | | |

## Then the fixed costs

| | Amount |
|---|---|
| Total margin from above | |
| Minus total fixed costs (week 20) | |
| **PROFIT SO FAR** | |

## Check it against the box

Count the money box, take away the startup fund you owe and the float you started with. You should get the same profit.

______________________________

## Closing inventory

Count what came home. Opening minus closing should equal what you sold. If it does not, your tally missed some.

| | Opened with | Came home with | So you sold | Tally said |
|---|---|---|---|---|
| Product 1 | | | | |
| Product 2 | | | | |

## Average sale

Total money in divided by number of customers. What is it? ________

▶ **ADD TO YOUR NUMBERS FILE**

Page 18 of your Numbers File: the by-product table, profit so far, the money box check and both inventory counts.

💬 **TALK ABOUT IT**

Tell an adult your money-in number, then tell them your profit number.

Ask which one they would have guessed was closer to the other.

**STUCK? TRY THIS**

Do one product row all the way across before starting the next. Mixing the two is how the totals stop agreeing with the money box.

★ **CHALLENGE ZONE**

Your two inventory counts and your tally probably disagree slightly.

Write down three real reasons that happens at a busy stall. Only one of them is someone being careless.

# Forecast Against Actual

Open week 23. Put the two numbers side by side and do not flinch.

**IN CLASS THIS WEEK**
Subtracting decimals; interpreting differences (5.NBT.7)

**MONEY WARM-UP · 2 MINUTES**
You forecast 25 and sold 18. What is the difference, and is it over or under? ________

**WORDS TO KEEP**

**variance** The gap between what you forecast and what actually happened.

**◂ YOU WILL NEED, FROM YOUR NUMBERS FILE**
- page 11: the forecast you wrote in pen in week 23
- page 18: what actually happened


## The comparison

| | Forecast | Actual | Difference | Over or under? |
|---|---|---|---|---|
| Product 1 sold | | | | |
| Product 2 sold | | | | |
| Money in | | | | |
| Profit | | | | |

### Which one was most wrong?

Not the biggest difference in dollars. The biggest difference compared with what you predicted.

| Row | Difference | As a share of the forecast |
|---|---|---|
| Product 1 | | |
| Product 2 | | |
| Money in | | |

### Why

Take the row you were most wrong about and write the most likely reason. Then write a second, different reason. Both may be true.

______________________________

______________________________

▶ **ADD TO YOUR NUMBERS FILE**

Page 19 of your Numbers File: the forecast against actual table, filled in, and the two reasons for the row you were most wrong about.

💬 **TALK ABOUT IT**

Show an adult your forecast and your actual, side by side.

Ask whether they have ever had to say out loud that a plan came in under. What happened next?

**STUCK? TRY THIS**

Being wrong is the normal result and it is not the interesting part. Which direction, and by how much compared with what you predicted, is the interesting part.

★ **CHALLENGE ZONE**

Suppose you had forecast perfectly. Would that have been skill, or luck?

Write down what a forecaster could do to be reliably close, and note that doing it requires having been wrong at least once first.

WEEK 32 MONEY

# What the Numbers Say

Slice the session four ways. One of the slices is telling you something.

**IN CLASS THIS WEEK**
Interpreting data; fractions of a quantity (5.MD.2, 5.NF.6)

**MONEY WARM-UP · 2 MINUTES**
Of 27 sales, 18 were product 1. What fraction is that, in simplest form? ________

**WORDS TO KEEP**

**segment** One slice of the day, or one product, looked at on its own.

**◂ YOU WILL NEED, FROM YOUR NUMBERS FILE**

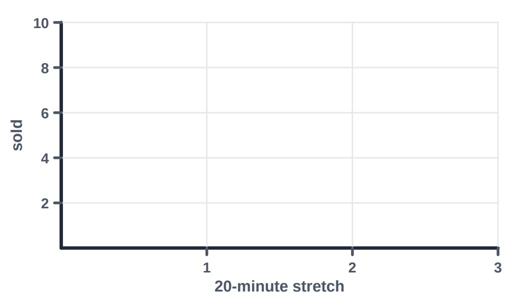
- page 17: the tally and the traffic count
- page 18: the by-product table

## Four slices

| Slice | What you look at | What you found |
|---|---|---|
| By product | which one sold, which one paid | |
| By time | which stretch was busy | |
| By customer | average sale, and who bought | |
| By price | did anyone hesitate at a number? | |

## Plot the session

Your three time stretches from week 29, plotted.



## The one finding

Out of everything above, write the single most useful thing you learned. It should be a sentence with a number in it.

Weak: 'product 2 did not sell well'. Strong: 'product 2 was a third of what we brought and a tenth of what we sold'.

## What you cannot tell

Name one thing your numbers do **not** tell you, however hard you look at them.

**▶ ADD TO YOUR NUMBERS FILE**

Page 20 of your Numbers File: all four slices, the plotted session, and your one finding written as a sentence with a number in it.

**💬 TALK ABOUT IT**

Show an adult your four slices and ask which one they find most interesting.

Ask what they would want to know next.

**STUCK? TRY THIS**

Start with the by-product slice. It is usually the one with the clearest story, and it points at what to change in week 33.

**★ CHALLENGE ZONE**

Two of your slices might disagree with each other.

If they do, write down what could make both true at once. If they agree, write down what would have to have happened for them to disagree.

# Change One Thing

One change, written down with what you expect it to do. Only one.

**IN CLASS THIS WEEK**
Multi-step problems; predicting with decimals (5.NBT.7)

**MONEY WARM-UP · 2 MINUTES**
You sell 20 at $4.00 and your margin is $1.80 each. If you drop the price to $3.50 and sell 6 more, do you make more or less? ________

**WORDS TO KEEP**

**one change** Change a single thing, so you know what caused the difference.

**◂ YOU WILL NEED, FROM YOUR NUMBERS FILE**

- page 20: your one finding from week 32

## Why only one

Change three things and session two will be different. You will have no idea which change did it.

Change one, and whatever happens, you know what caused it. This is exactly how a scientist runs an experiment and it is exactly how a business finds out what works.

## The candidates

| Change | What you would expect | How would you know it worked? |
|---|---|---|
| Lower a price | | |
| Raise a price | | |
| Bring more of one product | | |
| Change the sign | | |
| Move where you stand | | |

## Your one change

| | Your answer |
|---|---|
| We are changing | |
| Because our week 32 finding said | |
| We expect this to happen | |
| We will know it worked if | |
| Everything else stays | the same |

## Forecast session two

New forecast, in pen, before you sell anything.

| | How many | Money in | Profit |
|---|---|---|---|
| Product 1 | | | |
| Product 2 | | | |
| **TOTAL** | | | |

**▶ ADD TO YOUR NUMBERS FILE**

Page 21 of your Numbers File: your one change, why you picked it, what you expect, and your session two forecast in pen.

**💬 TALK ABOUT IT**

Tell an adult your one change and what you expect.

Ask them to guess what will actually happen. Write their guess down next to yours.

**STUCK? TRY THIS**

Pick the change your week 32 finding points at. A change you cannot connect to something you measured is a guess, and you already have a session's worth of measurements.

**★ CHALLENGE ZONE**

Suppose session two goes better, but the weather was also nicer and more people came.

Write down how you could still tell whether your change worked. Hint: sales per person who walked past is a fairer measure than sales.

WEEK 34 RUNNING IT

# Session Two

Same table, same page, one thing different. Take this to the stall.

**IN CLASS THIS WEEK**
Recording and interpreting data (5.MD.2)

**MONEY WARM-UP · 2 MINUTES**
Write your session two forecast total from week 33 here: ________

**WORDS TO KEEP**

**controlled** One thing changed, everything else the same, so the result means something.

## Tally as it happens

| | First half | Second half | Total sold |
|---|---|---|---|
| Product 1 | | | |
| Product 2 | | | |
| **BOTH** | | | |

## Traffic again

Count it the same way as session one, or the two sessions cannot be compared.

| Time | People about | Sales in that stretch |
|---|---|---|
| First 20 min | | |
| Second 20 min | | |
| Third 20 min | | |

## Did anyone react to the change?

Write down anything a customer said that was about the thing you changed.

______________________________

## What people said

| What they said, in their words | Bought? |
|---|---|
| 1. | |
| 2. | |
| 3. | |
| 4. | |

## Money in and closing count

| | Amount |
|---|---|
| Money in | |
| Closing inventory, product 1 | |
| Closing inventory, product 2 | |

**▶ ADD TO YOUR NUMBERS FILE**

Page 22 of your Numbers File: this whole page, filled in on the day.

**TALK ABOUT IT**

Before adding anything up, write one sentence on whether it felt different.

Then ask somebody who watched both sessions what they noticed.

**STUCK? TRY THIS**

Count the traffic the same way you did in session one, even if it feels fussy. A measurement taken differently cannot be compared with the first one.

**★ CHALLENGE ZONE**

Your change might have worked and the session might still have been worse.

Write down how both of those could be true at the same time. Naming that possibility now stops you drawing the wrong conclusion next week.

WEEK 35 REVIEW

# Did It Work?

Two sessions, one change, and an honest answer about what it did.

**IN CLASS THIS WEEK**
Comparing and subtracting decimals (5.NBT.7, 5.NF.2)

**MONEY WARM-UP · 2 MINUTES**
Session one: 27 sales from 60 people. Session two: 31 from 80. Which had the better rate?

______

**WORDS TO KEEP**

**comparison** Two sessions measured the same way, so the numbers mean something.

**◂ YOU WILL NEED, FROM YOUR NUMBERS FILE**

- page 17 and page 22: both session tallies
- page 21: the change you made and what you expected

## Side by side

| | Session 1 | Session 2 | Difference |
|---|---|---|---|
| Product 1 sold | | | |
| Product 2 sold | | | |
| Money in | | | |
| Profit | | | |
| People who came past | | | |

## The fair comparison

More people came to one session than the other, so raw sales are not a fair test. Work out sales per person who walked past.

| | Sales | People | Sales per person |
|---|---|---|---|
| Session 1 | | | |
| Session 2 | | | |

## Your verdict

| Question | Your answer |
|---|---|
| What did you change? | |
| What did you expect? | |
| What actually happened? | |
| **Did the change work?** | |
| How sure are you, and why? | |

## The honest option

“We cannot tell” is a real answer and sometimes the right one. If that is yours, say what would have made it clearer.

---

**▶ ADD TO YOUR NUMBERS FILE**

Page 23 of your Numbers File: the two sessions side by side, the sales-per-person comparison, and your verdict.

**💬 TALK ABOUT IT**

Tell an adult what you changed and what happened.

Ask them whether they believe the change caused it, and what would convince them.

**STUCK? TRY THIS**

Compare sales per person, not sales. If twice as many people came to session two, twice the sales is exactly the same performance.

**★ CHALLENGE ZONE**

One session is not proof. A change that seems to work once may just have been a good day.

Write down how many sessions you would want before you believed it, and why. There is no single right number, but there is a right kind of reasoning.

WEEK 36 REVIEW

# Your Numbers Story

A year of measurements, told as one page, with the numbers still attached.

**IN CLASS THIS WEEK**
Review; writing an opinion supported by facts (W.5.1)

**MONEY WARM-UP · 2 MINUTES**
Look back at week 1. What did you think a business's hardest number was? ________

**WORDS TO KEEP**

**track record** What you did, written down at the time. It is worth more than a memory of it.

**◂ YOU WILL NEED, FROM YOUR NUMBERS FILE**

- all twenty-three pages

## The year, in figures

Every line is already in your Numbers File.

| | Your answer |
|---|---|
| Our problem and our customer (week 14) | |
| What we made, and what we cut (weeks 15, 18) | |
| All-in cost of one (week 19) | |
| Our prices and margins (week 21) | |
| Break even, in items (week 22) | |
| Forecast, and what actually happened (weeks 23, 31) | |
| What we changed, and whether it worked (weeks 33, 35) | |
| Total profit across both sessions | |
| Split between partners | |

## Three things you know now

Specific ones, each with a number in it.

1. ______________________________
2. ______________________________
3. ______________________________

## The number that surprised you most

Which single figure was furthest from what you expected in September, and what did it change about how you think?

______________________________

______________________________

## For next year's class

One sentence you wish someone had told you in week 1.

______________________________

**▶ ADD TO YOUR NUMBERS FILE**

Page 24, the last page of your Numbers File: your numbers story. Keep the file. It is the measured record of a real business you ran.

**💬 TALK ABOUT IT**

Read your whole story out loud to somebody at home.

Ask them which number they found most surprising.

**STUCK? TRY THIS**

Read your Numbers File from page 1. The forecast you wrote in week 23 will look different now that you know what happened, and that is the whole point of having written it in pen.

**★ CHALLENGE ZONE**

A business that ran twice knows something a business that ran once does not.

Write down what that is, in one sentence. Then say what a third session would have told you that two could not.

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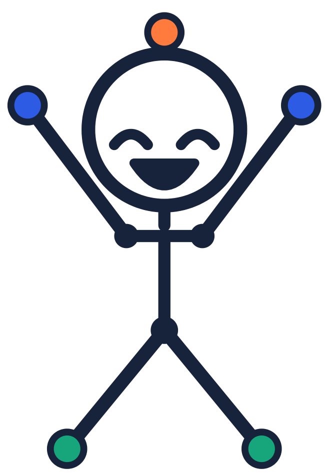
Grade 5 · The Numbers Year

# Certificate of Completion

This is to certify that

completed thirty-six weeks of enterprise work: measured the arithmetic of a business, built a line of products with a partner, wrote a forecast before anything was sold, sold twice to real customers, changed one thing in between, and worked out honestly whether that change made any difference.



Teacher

Date

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